

Raghav Senthil Kumar

623-332-2168 | raghavs9@illinois.edu | linkedin.com/in/raghav | github.com/raghav

EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. in Computer Science and Statistics

Expected: May 2029

- **Coursework:** Computer Science I (Java), Computer Science II (C++), Data Science & ML (Python), Calculus III
- **Activities:** Disruption Lab, SIGRobotics, AI Alignment, BuildIllinois, Innovation LLC, Agentic AI

EXPERIENCE

Founding Software Engineer | Clusion

September 2025 – January 2026

- Collaborated with a multidisciplinary *Agile* team of **10 software developers** to jointly define software requirements, architecture and design for an *AWS*-hosted SaaS platform for debate research, drawing **10K+ site visitors**
- Designed and implemented back end workflows and *REST APIs* in a Python web framework (*FastAPI*) that index **600K+ debate-evidence documents** on *Elasticsearch* and return search results to users in **under 1 second**
- Wrote a text-retrieval algorithm from scratch that validates source URLs and filters statistical outliers with **IQR**
- Built back end de-duplication and recency-ranking workflows, flagging **30%** of documents and improving the UX

Product Development Intern | Jetson

January 2025 – February 2025

- Rapidly prototyped and fully developed a new onboarding user interface concept in *Figma* with a focus on overall user experience, cutting the flow from **12 to 5 pages** and validating it through **A/B testing** with 30 users
- Ran **7 rounds of UI/UX testing** on 3 new UI features for **100K+ users**, iterating on each interface concept
- Presented a **5-page report** to the CEO turning test results into UX requirements and design recommendations

Data Analytics Intern | STEM E

June 2024 – July 2024

- Wrote *Python (pandas)* scripts and built *Power BI* dashboards to analyze 3 years of data (**200+ TikTok posts**)
- Found that “science fun facts” posts drove 69.3% of views while making up only 7% of all published content
- Worked closely with the Executive Director to turn it into a new strategy, lifting engagement **24% (8K+ views)**

PROJECTS

Chronos | [Live Website](#) | [Source Code](#)

December 2025

- Designed, wrote and tested meeting management software from scratch, building *React* UI components and back end workflows on a *Firebase* database that auto-book meetings for groups of **30+ people** onto Google Calendar
- Implemented a **5-step Python** back end workflow that scans calendars, ranks open slots and emails confirmations
- Validated the workflow from availability scan to booking confirmation with **14 unit and integration tests**
- Developed a secure lobby management feature where hosts authenticate invitees and queue follow-up meetings
- Secured licenses with **8 startups** and **2 LOIs (\$3.5K)**. **Winner, 2025 Congressional App Challenge**

SchoolPulse AI | [Live Website](#) | [Source Code](#)

June 2026

- Built a network of **3 edge AI hardware/software tools** tracking energy, water and food use across a school
- Architected a *Python FastAPI* back end whose workflows feed sensor data to **4 Random Forest** anomaly models
- Developed a *Next.js* dashboard user interface where a **Gemma Voice LLM** delivers actionable steps that help schools monitor, understand and manage their energy, water and food consumption as live sensor data arrives
- Led a multidisciplinary team of 5 from requirements through final demo. **USAII Global Hackathon Finalist**

EcoBin | [Research Paper](#) | [Source Code](#)

March 2026 – May 2026

- Rapidly prototyped and tested a hardware/software smart bin from scratch on an *Arduino Uno* microcontroller, writing the embedded control software that sorts each waste item into its disposal pathway in **under 2 seconds**
- Designed a two-stage *EfficientNetV2-S* neural network architecture (**4.39M** parameters) in *Python (PyTorch)* and trained it on **15K+** labeled images, reaching **96.13% accuracy** when sorting waste across 4 disposal categories
- Built a synthetic data workflow (*U2-Net* segmentation, alpha-compositing) producing **9K** contamination images
- Published the work as an *arXiv* paper with 2 open *Kaggle* datasets. **2nd at Arizona State Science Fair**

SKILLS

Programming Languages: Python, SQL, JavaScript, HTML/CSS, Java, C++

Frameworks/Platforms: FastAPI, Flask, PostgreSQL, MySQL, React, Next.js, Firebase, Node.js, AWS, React Flow

DevTools: Figma, Git, GitHub, Docker, CI/CD, VS Code, Arduino IDE, Kaggle, Claude Code, Codex, Bun

Concepts/Practices: Full-Stack Development, Unit Testing, REST APIs, API Development, Authentication, Agile